



— CAPABILITY STATEMENT

SCIENCE SYSTEMS AND APPLICATIONS, INC.

SCIENCE AND TECHNOLOGY WITH PASSION

UEI PJ2SP76D3HJ3

CAGE 5S009

NAICS 541715 (small) / 541330

WOSB · ITAR Registered

ISO 9001 · AS9100 · CMMI L3

Top Secret Facility Clearance

Turning science into mission advantage. SSAI is a leading science and engineering partner to U.S. agencies, with expertise in Earth and space data, AI/ML analytics, instrument engineering, and secure IT solutions. For over four decades, we have transformed complex science into actionable insights that drive national missions. Engineering excellence trusted on NASA's flagship missions, now enabling defense and national security.

1977

FOUNDED · 45+ YEARS

575+

SCIENTISTS,
ENGINEERS & IT
PROFESSIONALS

160+

MISSIONS SERVED

100+

MAJOR INSTRUMENTS,
MODELS & MISSIONS

10,000+

PUBLICATIONS & DATA
PRODUCTS

3

WHOLLY-OWNED
SUBSIDIARIES

CORE CAPABILITIES

SCIENCE AND TECHNOLOGY SOLUTIONS FOR EVERY MISSION SPACE.

Five capability horizontals span the full arc from research and instrument engineering to manufacturing, geospatial intelligence, and secure digital defense.

- 01 SCIENCE, TECHNOLOGY & ENGINEERING SERVICES**
Systems engineering, requirements and V&V, avionics design, software engineering (CMMI L3), FPGA design, EEE parts processing, system integration and testing, and manufacturing engineering. AS9100 and ITAR, J-STD-001F workmanship. Proven on NASA flagships, now enabling defense and national security.
- 02 SPACE OPERATIONS**
Deep space and space weather. We operate NASA's GEOS Data Assimilation System (millions of observations daily), global atmospheric reanalysis, and 24/7 operational modeling for NOAA and NASA, plus deep-space instrument development and calibration, trajectory design, and mission operations from pre-launch through on-orbit.
- 03 SATELLITE & SPACE SYSTEMS MANUFACTURING**
A 32,000 sq ft electronics manufacturing facility and a 10,000 sq ft machine shop under two miles from NASA Goddard. ESD-protected floor, three clean rooms, cleared personnel, Mil-spec CNC machining, and certified aerospace machinists. End-to-end, prototyping to production.
- 04 ENVIRONMENTAL INTELLIGENCE**
Geospatial and remote sensing. We transform raw data into actionable intelligence: data assimilation, remote sensing, oceanography, hydrology, biosphere modeling, and weather and space-weather forecasting across instruments including OMI, OMPS, MODIS, VIIRS, MISR, ICESat-2, AVIRIS-NG, TROPOMI, and GEMS.
- 05 CYBERSECURITY & DIGITAL DEFENSE**
Advanced IT, cybersecurity (zero-trust architectures), and AI/ML built for national security: custom AI for digital twins, anomaly detection, and cross-agency support, anchored by CREST (Coupled Reusable Earth System Tensor), an AI-enabled Python digital-twin framework. Focused on operational dominance.

DIFFERENTIATORS

WHY SSAI.

- **Beside NASA Goddard, in the D.C. defense corridor.** Headquartered minutes from NASA Goddard and the Washington defense community, embedded where missions are engineered and flown.
- **45+ years of mission heritage.** Founded 1977, with 575+ scientists, engineers, and IT professionals delivering across 160+ missions.
- **A single, trusted source.** Three wholly-owned subsidiaries bring electronics manufacturing, cybersecurity engineering, and precision machining in house.
- **Cleared and certified.** Top Secret Facility Clearance and ITAR registration, paired with appraised process maturity and aerospace-grade quality systems.

AS9100 Aerospace Quality

CMMI L3 Development

ISO 9001:2015

Top Secret FCL

ITAR Registered

SUBSIDIARIES

MISSION-READY, IN HOUSE.

A single, trusted source for mission-ready capability.

ADVANCED
MISSION
PARTNERS

Electronics manufacturing, engineering services, EEE expertise, and precision manufacturing.

ELUCIDATION
CONCEPTS,
LLC

Cybersecurity solutions, systems engineering, system integration testing, and test & evaluation.

SIGMA
ENGINEERING
&
FABRICATION,
INC.

Precision machining, manufacturing, material expertise, and quality at speed.

PAST PERFORMANCE

DELIVERING MISSION-CRITICAL SCIENCE AND ENGINEERING.

ESES III	NASA Goddard, Engineering Technology Directorate (CPAF). Study, design, development, fabrication, integration, testing, verification, and operations of spaceflight, airborne, and ground hardware and software. 11/2018 to 04/2025.	\$685M SSAI TO CONFIRM
HBG	Hydrospheres, Biospheres and Geophysics. NASA Goddard, Earth Sciences Division (CPFF). Scientific research services across space, airborne, in-situ, and laboratory observations. 04/2020 to 11/2025.	\$425M SSAI TO CONFIRM
STARSS III	Science, Technology and Research Support Services. NASA Langley, Science Directorate (CPFF). Earth atmosphere and climate, extending to other planetary atmospheres. 12/2016 to 05/2023.	\$307M SSAI TO CONFIRM
SAMDA	Support for Atmospheric Modeling and Data Assimilation. NASA Goddard, Global Modeling and Assimilation Office (CPFF). Research across the Earth Sciences Division. 03/2017 to 11/2025.	\$298M SSAI TO CONFIRM
HBS	Hydrospheric and Biospheric Sciences. NASA Goddard, Earth Sciences Division (CPFF). 12/2014 to 05/2020.	\$249M SSAI TO CONFIRM

Flagship missions: OCI / PACE · Roman Space Telescope · Lucy · Mars Sample Return / CCRS · Dragonfly · GeoXO · DAVINCI+ · TSIS-2 · JWST.

MARKETS SERVED

TRUSTED ACROSS SIX CUSTOMER VERTICALS.

DEFENSE &
NATIONAL
SECURITY

DoD and defense labs.

FEDERAL
CIVILIAN

NASA, NOAA, USGS,
EPA, USDA, GSA,
Commerce, Interior.

INTELLIGENCE
COMMUNITY

Secure mission support.

STATE & LOCAL

LA World Airports, LA
Unified School District.

COMMERCIAL

Industry and partners.

OCONUS /
INTERNATIONAL

World Bank and global
programs.

Customer set: NASA (Goddard, Langley), NOAA, USGS, EPA, USDA, GSA, Department of Commerce, Department of the Interior, DoD and defense labs, LAWA, LAUSD, and the World Bank.

CORPORATE DATA

THE ESSENTIALS.

UEI	PJ2SP76D3HJ3 CONFIRM	CAGE	5S009 CONFIRM
NAICS	541715 (small) / 541330	BUSINESS STATUS	WOSB, SBA 2024 CONFIRM
GSA MAS	47QRAA23D0014 CONFIRM	IDIQ VEHICLE	OASIS Small Business CONFIRM
QUALITY & MATURITY	ISO 9001:2015, AS9100, CMMI L3	SECURITY	Top Secret FCL, ITAR Registered
FOUNDED	1977	HEADQUARTERS	Lanham, MD

BUSINESS DEVELOPMENT

BD@ssaihq.com
(301) 867-2000

HEADQUARTERS

The Aerospace Building
10210 Greenbelt Rd, Suite 600
Lanham, MD 20706

LOCATION ADVANTAGE

Minutes from NASA Goddard, in the
Washington D.C. defense corridor.

Procurement identifiers and contract values marked "confirm" / "SSAI to confirm" are pending SSAI sign-off before external release.